



Term End Examination

May/June 2024

CET2012B - Data Engineering Concepts

Question Paper ID: 036388

Faculty/School	Engineering and Technology	Term	Semester V
Program	TY B.Tech CSE	Duration	2 Hours 30 Minutes
Specialization	-	Max. Marks	70

Section - 1 (7 X 10 Marks)

Answer any 7 questions

1	A. Elaborate the characteristics of semi-structured data with appropriate example. B. With the help of suitable examples, explain the concept of nominal and ordinal attributes.	10 marks	CO1, CO2	Understanding
2	A. Elaborate various types of attribute subset selection. B. With the help of real world example, explain the concept of data discretization.	10 marks	CO2	Understanding
3	A. Design a star schema for a healthcare provider's data warehouse. The fact table could contain patient count, and dimensions include patients, doctors, diagnoses, procedures, and time. Justify the type of measures for patient count with explanation. B. Consider a dataset with the following values for a specific feature: [15,22,18,30,12] Find the Min-Max normalization for each value in the dataset.	10 marks	CO2, CO3	Applying
4	Design of snow flake schema is different from fast constellation schema. Identify the difference by drawing a diagram for Sales Analysis as Fact Table and dimensions as time, branch, location and item for both types of schema.	10 marks	CO3	Applying
5	A. Data is stored in a data cube for an electronic company based on products, date and country. Draw a data cube diagram with three dimensions and convert it into data cuboid. B. "Compare and contrast the functionalities of data warehouses and operational databases. How does the distinct design of each type contribute to their respective roles in managing data?"	10 marks	CO3	Applying

6	Construct Frequent Pattern (FP) tree based on following Transactions: Minimum Support: 2 <table><tr><th>Transaction ID</th><th>Items</th></tr><tr><td>1</td><td>B C D A</td></tr><tr><td>2</td><td>E D C</td></tr><tr><td>3</td><td>A B</td></tr><tr><td>4</td><td>A C D</td></tr><tr><td>5</td><td>F G D B</td></tr></table>	Transaction ID	Items	1	B C D A	2	E D C	3	A B	4	A C D	5	F G D B	10 marks	CO3	Applying																								
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7	A. With proper design of data warehouse for an organization, elaborate the benefits that the business analyst gain. B. Elucidate the top-down data warehouse design process with diagram. Identify the benefits of this approach.	10 marks	CO3	Understanding																																				
8	Suppose we have a dataset of students and whether they passed or failed based on two features: "Study Hours" and "Attendance." <table><tr><th>Student</th><th>Study Hours</th><th>Attendance</th><th>Passed</th></tr><tr><td>1</td><td>2</td><td>Low</td><td>No</td></tr><tr><td>2</td><td>3</td><td>High</td><td>No</td></tr><tr><td>3</td><td>5</td><td>High</td><td>Yes</td></tr><tr><td>4</td><td>1</td><td>Low</td><td>No</td></tr><tr><td>5</td><td>4</td><td>High</td><td>Yes</td></tr><tr><td>6</td><td>2</td><td>Low</td><td>No</td></tr><tr><td>7</td><td>6</td><td>High</td><td>Yes</td></tr><tr><td>8</td><td>3</td><td>Low</td><td>No</td></tr></table> Calculate the entropy for the whole dataset and information gain for Study Hours and Attendance.	Student	Study Hours	Attendance	Passed	1	2	Low	No	2	3	High	No	3	5	High	Yes	4	1	Low	No	5	4	High	Yes	6	2	Low	No	7	6	High	Yes	8	3	Low	No	10 marks	CO4	Analysing
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9	Average stock prices of two financial organizations: Westpac and Commonwealth are given in the following Table for five months. If the stocks are affected by same industry trends, determine whether the price will rise or fall together. Apply covariance measure. <table><tr><th>Month</th><th>Westpac</th><th>Commonwealth</th></tr><tr><td>January</td><td>2</td><td>5</td></tr><tr><td>February</td><td>3</td><td>8</td></tr><tr><td>March</td><td>5</td><td>10</td></tr><tr><td>April</td><td>4</td><td>11</td></tr><tr><td>May</td><td>6</td><td>14</td></tr></table>	Month	Westpac	Commonwealth	January	2	5	February	3	8	March	5	10	April	4	11	May	6	14	10 marks	CO2	Applying																		
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10	A. Explain the components of a decision tree, including root node, internal nodes, branch and leaves. Elaborate the process of splitting of decision tree. B. Explain k-means clustering algorithm. Detail any one application of this algorithm.	10 marks	CO4	Applying
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END OF QUESTION PAPER

